Architecture note: demo layer to production equivalent

Concept demo with sample data. One page: each demo layer and its production equivalent.

Layer	In this demo	In production
Identity	Role picker, HMAC-signed httpOnly cookie	SSO (OIDC/SAML), MFA, SCIM-managed roles
Authorisation	One permission map (src/lib/permissions.ts) drives code and the Governance matrix; PostgreSQL RLS enforces rows	Same map, plus column-level views and policy tests in CI
Database	PGlite locally, Neon PostgreSQL on Vercel, same Drizzle schema and RLS SQL	Managed PostgreSQL with PITR, read replicas, RLS unchanged
Data	Deterministic synthetic seed (30 assets, 400 sales, 36-month series)	Ingestion from registries, brokers, bank and property-management systems, with lineage
Assumptions	Versioned, SHA-256-hashed sets in assumption_sets	Same, with approval workflow and release notes
Audit	Hash-chained audit_events, append-only by policy	Same chain, written by a SECURITY DEFINER function, anchored externally
Valuation	Deterministic TypeScript hedonic model, versioned hashed assumptions, comparable-sales cross-check, two-person confirmation	Same engine and register, calibrated on real transactions, with model risk governance and back-testing
AI (Horus)	One server-side gateway; the model calls deterministic tools under the user's row-level security; every claim needs a citation id a tool returned and every number must come from a tool; scripted fallback	Same gateway with evals, prompt/version registry, cost controls, streaming and server-side refusal fallbacks
Scenarios	Pure TypeScript engine, sample coefficients in a versioned hashed register, saved with input, set version, data hash and result hash	Same engine and register, coefficients calibrated and approved by a risk committee, results stored as immutable runs
Documents	Rule-based reader for three bundled sample PDFs, confidence per field, human confirmation before any record changes	OCR and model extraction with a review queue, document store, retention and legal-hold rules
Investor pack	react-pdf server-side, English and Arabic, content hash bound to the audit chain head, issue register	Same, plus signed PDFs, a branded template system and delivery through a secure portal
Alerts and approvals	Pure rule engine (ALERT_ASSUMPTIONS), snooze needs a reason, one approvals queue over valuations, documents, update drafts and role changes, separation of duties	Same rules as configuration, notification channels, escalation and SLAs
Chain verification	Verify chain recomputes every SHA-256 link and names the first broken event; admin-only labelled tamper demo	Scheduled verification, external anchoring, no tamper action
Live feed	Simulated events over SSE, role-filtered, never stored	Events from the ledger and property-management systems via a queue
Database drivers	PGlite, Neon serverless, or node-postgres chosen from the URL; provisioning in one transaction under an advisory lock	Migrations through CI, no runtime provisioning
i18n	next-intl, /en and /ar, true RTL, Western digits	Same, with a managed translation workflow
Hosting	Vercel	Vercel or a regional cloud, depending on data-residency requirements

the model never does arithmetic or decides; every number links to source rows and an assumption set; synthetic data is labelled wherever it appears.